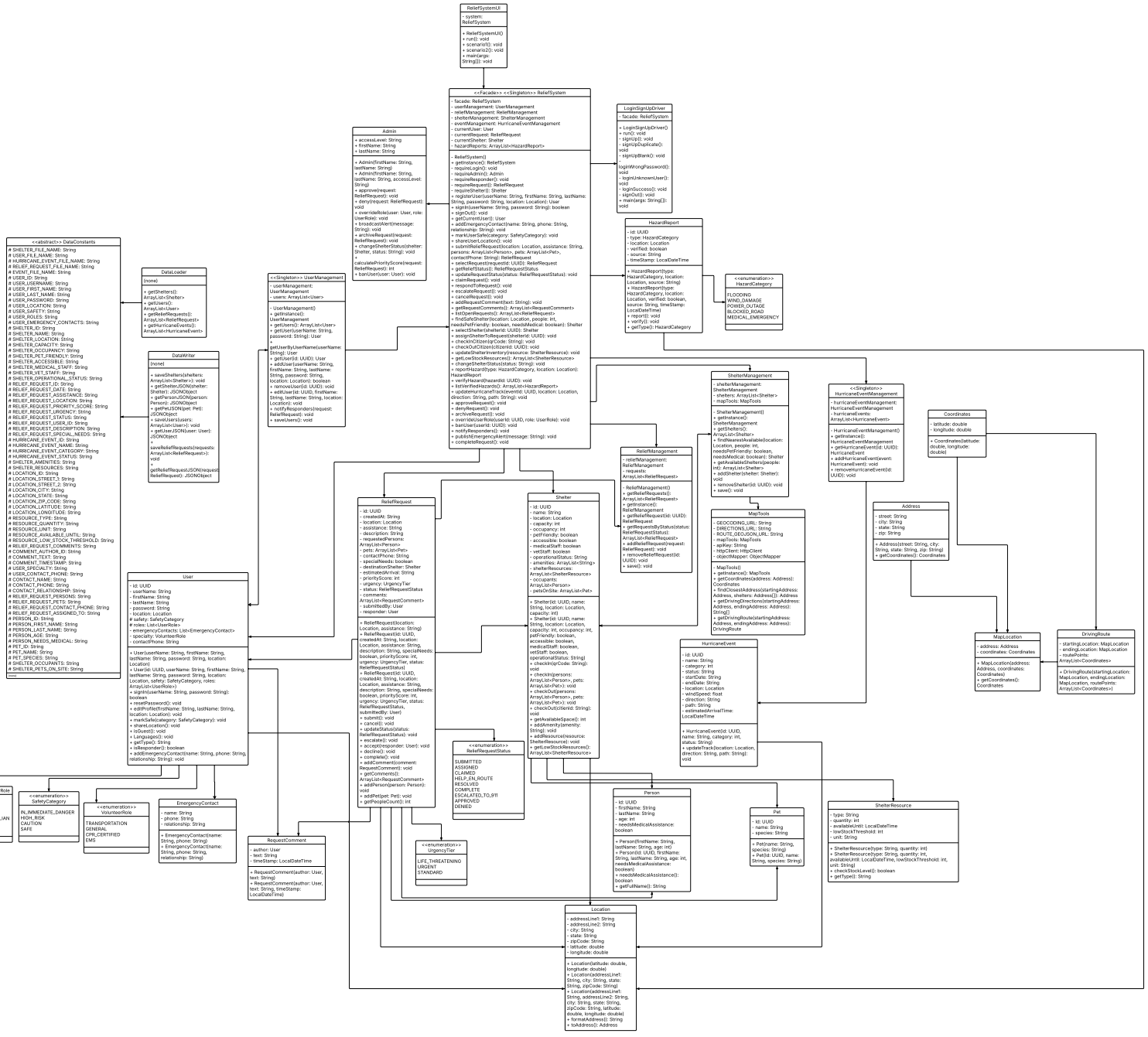

ReliefSystemUI
- system: ReliefSystem
+ ReliefSystemUI()
+ run(): void
+ scenario0(): void
+ scenario2(): void
+ main(args: String[]): void
<<Facade>> <<Singleton>> ReliefSystem
- facade: ReliefSystem
- userManagement: UserManagement
- reliefManagement: ReliefManagement
- shelterManagement: ShelterManagement
- eventManagement: HurricaneEventManagement
- currentUser: User
- currentRequest: ReliefRequest
- currentShelter: Shelter
- hazardReports: ArrayList<HazardReport>
- ReliefSystem()
+ getInstance(): ReliefSystem
- requireLogin(): void
- requireAdmin(): Admin
- requireResponder(): void
- requireRequest(): ReliefRequest
- requireShelter(): Shelter
+ registerUser(userName: String, firstName: String, lastName: String, password: String, location: Location): User
+ signIn(userName: String, password: String): boolean
+ signOut(): void
+ getCurrentUser(): User
+ addEmergencyContact(name: String, phone: String, relationship: String): void
+ markUserSafe(category: SafetyCategory): void
+ shareUserLocation(): void
+ submitReliefRequest(location: Location, assistance: String, persons: ArrayList<Person>, pets: ArrayList<Pet>, contactPhone: String): ReliefRequest
+ selectRequest(requestId: UUID): ReliefRequest
+ getReliefStatus(): ReliefRequestStatus
+ updateRequestStatus(status: ReliefRequestStatus): void
+ claimRequest(): void
+ respondToRequest(): void
+ escalateRequest(): void
+ cancelRequest(): void
+ addRequestComment(text: String): void
+ getRequestComments(): ArrayList<RequestComment>
+ listOpenRequests(): ArrayList<ReliefRequest>
+ findSafeShelter(location: Location, people: int, needsPetFriendly: boolean, needsMedical: boolean): Shelter
+ selectShelter(shelterId: UUID): Shelter
+ assignShelterToRequest(shelterId: UUID): void
+ checkInCitizen(qrCode: String): void
+ checkOutCitizen(citizenId: UUID): void
+ updateShelterInventory(resource: ShelterResource): void
+ getLowStockResources(): ArrayList<ShelterResource>
+ changeShelterStatus(status: String): void
+ reportHazard(type: HazardCategory, location: Location): HazardReport
+ verifyHazard(hazardId: UUID): void
+ listVerifiedHazards(): ArrayList<HazardReport>
+ updateHurricaneTrack(eventId: UUID, location: Location, direction: String, path: String): void
+ approveRequest(): void
+ denyRequest(): void
+ archiveRequest(): void
+ overrideUserRole(userId: UUID, role: UserRole): void
+ banUser(userId: UUID): void
+ notifyResponders(): void
+ publishEmergencyAlert(message: String): void
+ completeRequest(): void
LoginSignUpDriver
- facade: ReliefSystem
+ LoginSignUpDriver()
+ run(): void
- signUp(): void
- signUpDuplicate(): void
- signUpBlank(): void
- loginWrongPassword(): void
- loginUnknownUser(): void
- loginSuccess(): void
- signOut(): void
+ main(args: String[]): void
Admin
+ accessLevel: String
+ firstName: String
+ lastName: String
+ Admin(firstName: String, lastName: String)
+ Admin(firstName: String, lastName: String, accessLevel: String)
+ approve(request: ReliefRequest): void
+ deny(request: ReliefRequest): void
+ overrideRole(user: User, role: UserRole): void
+ broadcastAlert(message: String): void
+ archiveRequest(request: ReliefRequest): void
+ changeShelterStatus(shelter: Shelter, status: String): void
+ calculatePriorityScore(request: ReliefRequest): int
+ banUser(user: User): void
HazardReport
- id: UUID
- type: HazardCategory
- location: Location
- verified: boolean
- source: String
- timeStamp: LocalDateTime
+ HazardReport(type: HazardCategory, location: Location, verified: boolean, source: String, timeStamp: LocalDateTime)
+ report(): void
+ verify(): void
+ getType(): HazardCategory
<<enumeration>>
HazardCategory
FLOODING
WIND_DAMAGE
POWER_OUTAGE
BLOCKED_ROAD
MEDICAL_EMERGENCY
<<abstract>> DataConstants
SHELTER_FILE_NAME: String
USER_FILE_NAME: String
HURRICANE_EVENT_FILE_NAME: String
RELIEF_REQUEST_FILE_NAME: String
EVENT_FILE_NAME: String
USER_ID: String
USER_USERNAME: String
USER_FIRST_NAME: String
USER_LAST_NAME: String
USER_PASSWORD: String
USER_LOCATION: String
USER_SAFETY: String
USER_ROLES: String
USER_EMERGENCY_CONTACTS: String
SHELTER_ID: String
SHELTER_NAME: String
SHELTER_LOCATION: String
SHELTER_CAPACITY: String
SHELTER_OCCUPANCY: String
SHELTER_PET_FRIENDLY: String
SHELTER_ACCESSIBLE: String
SHELTER_MEDICAL_STAFF: String
SHELTER_VET_STAFF: String
SHELTER_OPERATIONAL_STATUS: String
RELIEF_REQUEST_ID: String
RELIEF_REQUEST_DATE: String
RELIEF_REQUEST_ASSISTANCE: String
RELIEF_REQUEST_LOCATION: String
RELIEF_REQUEST_PRIORITY_SCORE: String
RELIEF_REQUEST_URGENCY: String
RELIEF_REQUEST_STATUS: String
RELIEF_REQUEST_USER_ID: String
RELIEF_REQUEST_DESCRIPTION: String
RELIEF_REQUEST_SPECIAL_NEEDS: String
HURRICANE_EVENT_ID: String
HURRICANE_EVENT_NAME: String
HURRICANE_EVENT_CATEGORY: String
HURRICANE_EVENT_STATUS: String
SHELTER_AMENITIES: String
SHELTER_RESOURCES: String
LOCATION_ID: String
LOCATION_STREET_1: String
LOCATION_STREET_2: String
LOCATION_CITY: String
LOCATION_STATE: String
LOCATION_ZIP_CODE: String
LOCATION_LATITUDE: String
LOCATION_LONGITUDE: String
RESOURCE_TYPE: String
RESOURCE_QUANTITY: String
RESOURCE_UNIT: String
RESOURCE_AVAILABLE_UNTIL: String
RESOURCE_LOW_STOCK_THRESHOLD: String
RELIEF_REQUEST_COMMENTS: String
COMMENT_AUTHOR_ID: String
COMMENT_TEXT: String
COMMENT_TIMESTAMP: String
USER_SPECIALTY: String
USER_CONTACT_PHONE: String
CONTACT_NAME: String
CONTACT_PHONE: String
CONTACT_RELATIONSHIP: String
RELIEF_REQUEST_PERSONS: String
RELIEF_REQUEST_PETS: String
RELIEF_REQUEST_CONTACT_PHONE: String
RELIEF_REQUEST_ASSIGNED_TO: String
PERSON_ID: String
PERSON_FIRST_NAME: String
PERSON_LAST_NAME: String
PERSON_AGE: String
PERSON_NEEDS_MEDICAL: String
PET_ID: String
PET_NAME: String
PET_SPECIES: String
SHELTER_OCCUPANTS: String
SHELTER_PETS_ON_SITE: String
+ test()
DataLoader
(none)
+ getShelters(): ArrayList<Shelter>
+ getUsers(): ArrayList<User>
+ getReliefRequests(): ArrayList<ReliefRequest>
+ getHurricaneEvents(): ArrayList<HurricaneEvent>
<<Singleton>> UserManagement
- userManagement: UserManagement
- users: ArrayList<User>
- UserManagement()
+ getInstance(): UserManagement
+ getUsers(): ArrayList<User>
+ getUser(userName: String, password: String): User
+ getUserByUserName(userName: String): User
+ getUser(id: UUID): User
+ addUser(userName: String, firstName: String, lastName: String, password: String, location: Location): boolean
+ removeUser(id: UUID): void
+ editUser(id: UUID, firstName: String, lastName: String, location: Location): void
+ notifyResponders(request: ReliefRequest): void
+ saveUsers(): void
DataWriter
(none)
+ saveShelters(shelters: ArrayList<Shelter>): void
+ getShelterJSON(shelter: Shelter): JSONObject
+ getPersonJSON(person: Person): JSONObject
+ getPetJSON(pet: Pet): JSONObject
+ saveUsers(users: ArrayList<User>): void
+ getUserJSON(user: User): JSONObject
+ saveReliefRequests(requests: ArrayList<ReliefRequest>): void
+ getReliefRequestJSON(request: ReliefRequest): JSONObject
ShelterManagement
- shelterManagement: ShelterManagement
- shelters: ArrayList<Shelter>
- mapTools: MapTools
- ShelterManagement()
+ getInstance(): ShelterManagement
+ getShelters(): ArrayList<Shelter>
+ findNearestAvailableLocation(Location, people: int, needsPetFriendly: boolean, needsMedical: boolean): Shelter
+ getAvailableShelters(people: int): ArrayList<Shelter>
+ addShelter(shelter: Shelter): void
+ removeShelter(id: UUID): void
+ save(): void
<<Singleton>>
HurricaneEventManagement
- hurricaneEventManagement: HurricaneEventManagement
- hurricaneEvents: ArrayList<HurricaneEvent>
+ HurricaneEventManagement()
+ getInstance(): HurricaneEventManagement
+ getHurricaneEvent(id: UUID): HurricaneEvent
+ addHurricaneEvent(event: HurricaneEvent): void
+ removeHurricaneEvent(id: UUID): void
Coordinates
- latitude: double
- longitude: double
+ Coordinates(latitude: double, longitude: double)
ReliefManagement
- reliefManagement: ReliefManagement
- requests: ArrayList<ReliefRequest>
- ReliefManagement()
+ getReliefRequests(): ArrayList<ReliefRequest>
+ getInstance(): ReliefManagement
+ getReliefRequest(id: UUID): ReliefRequest
+ getRequestsByStatus(status: ReliefRequestStatus): ArrayList<ReliefRequest>
+ addReliefRequest(request: ReliefRequest): void
+ removeReliefRequest(id: UUID): void
+ save(): void
MapTools
- GEOCODING_URL: String
- DIRECTIONS_URL: String
- ROUTE_OPTIMIZATION_URL: String
- mapTools: MapTools
- apiKey: String
- httpClient: HttpClient
- objectMapper: ObjectMapper
- MapTools()
+ getInstance(): MapTools
+ getCoordinates(address: Address): Coordinates
+ findClosestAddress(startingAddress: Address, shelters: Address[]): Address
+ getDrivingDirections(startingAddress: Address, endingAddress: Address): String[]
+ getDrivingRoute(startingAddress: Address, endingAddress: Address): DrivingRoute
Address
- street: String
- city: String
- state: String
- zip: String
+ Address(street: String, city: String, state: String, zip: String)
+ getCoordinates(): Coordinates
MapLocation
- address: Address
- coordinates: Coordinates
+ MapLocation(address: Address, coordinates: Coordinates)
+ getCoordinates(): Coordinates
DrivingRoute
- startingLocation: MapLocation
- endingLocation: MapLocation
- routePoints: ArrayList<Coordinates>
+ DrivingRoute(startingLocation: MapLocation, endingLocation: MapLocation, routePoints: ArrayList<Coordinates>)
ReliefRequest
- id: UUID
- createdAt: String
- location: Location
- assistance: String
- description: String
- requestedPersons: ArrayList<Person>
- pets: ArrayList<Pet>
- contactPhone: String
- specialNeeds: boolean
- destinationShelter: Shelter
- estimatedArrival: String
- priorityScore: int
- urgency: UrgencyTier
- status: ReliefRequestStatus
- comments: ArrayList<RequestComment>
- submittedBy: User
- responder: User
+ ReliefRequest(location: Location, assistance: String)
+ ReliefRequest(id: UUID, createdAt: String, location: Location, assistance: String, description: String, specialNeeds: boolean, priorityScore: int, urgency: UrgencyTier, status: ReliefRequestStatus)
+ ReliefRequest(id: UUID, createdAt: String, location: Location, assistance: String, description: String, specialNeeds: boolean, priorityScore: int, urgency: UrgencyTier, status: ReliefRequestStatus, submittedBy: User)
+ submit(): void
+ cancel(): void
+ updateStatus(status: ReliefRequestStatus): void
+ escalate(): void
+ accept(responder: User): void
+ decline(): void
+ complete(): void
+ addComment(comment: RequestComment): void
+ getComments(): ArrayList<RequestComment>
+ addPerson(person: Person): void
+ addPet(pet: Pet): void
+ getPeopleCount(): int
Shelter
- id: UUID
- name: String
- location: Location
- capacity: int
- occupancy: int
- petFriendly: boolean
- accessible: boolean
- medicalStaff: boolean
- vetStaff: boolean
- operationalStatus: String
- amenities: ArrayList<String>
- shelterResources: ArrayList<ShelterResource>
- occupants: ArrayList<Person>
- petsOnSite: ArrayList<Pet>
+ Shelter(id: UUID, name: String, location: Location, capacity: int)
+ Shelter(id: UUID, name: String, location: Location, capacity: int, occupancy: int, petFriendly: boolean, accessible: boolean, medicalStaff: boolean, vetStaff: boolean, operationalStatus: String)
+ checkIn(qrCode: String): void
+ checkIn(persons: ArrayList<Person>, pets: ArrayList<Pet>): void
+ checkOut(persons: ArrayList<Person>, pets: ArrayList<Pet>): void
+ checkOut(clientId: String): void
+ getAvailableSpace(): int
+ addAmenity(amenity: String): void
+ addResource(resource: ShelterResource): void
+ getLowStockResources(): ArrayList<ShelterResource>
User
- id: UUID
- userName: String
- firstName: String
- lastName: String
- password: String
- location: Location
safety: SafetyCategory
roles: List<UserRole>
- emergencyContacts: List<EmergencyContact>
- specialty: VolunteerRole
- contactPhone: String
+ User(userName: String, firstName: String, lastName: String, password: String, location: Location)
+ User(id: UUID, userName: String, firstName: String, lastName: String, password: String, location: Location, safety: SafetyCategory, roles: ArrayList<UserRole>)
+ signIn(userName: String, password: String): boolean
+ resetPassword(): void
+ editProfile(firstName: String, lastName: String, location: Location): void
+ markSafe(category: SafetyCategory): void
+ shareLocation(): void
+ isGuest(): void
+ Language(s): void
+ getType(): String
+ isResponder(): boolean
+ addEmergencyContact(name: String, phone: String, relationship: String): void
HurricaneEvent
- id: UUID
- name: String
- category: int
- status: String
- startDate: String
- endDate: String
- location: Location
- windSpeed: float
- direction: String
- path: String
- estimatedArrivalTime: LocalDateTime
+ HurricaneEvent(id: UUID, name: String, category: int, status: String)
+ updateTrack(location: Location, direction: String, path: String): void
<<enumeration>>
ReliefRequestStatus
SUBMITTED
ASSIGNED
CLAIMED
HELP_EN_ROUTE
RESOLVED
COMPLETE
ESCALATED_TO_911
APPROVED
DENIED
<<enumeration>> UserRole
ADMIN
SHELTER_MANAGER
VOLUNTEER
NON_VOLUNTEER_CIVILIAN
VICTIM
PROFESSIONAL
<<enumeration>>
SafetyCategory
IN_IMMEDIATE_DANGER
HIGH_RISK
CAUTION
SAFE
<<enumeration>>
VolunteerRole
TRANSPORTATION
GENERAL
CPR_CERTIFIED
EMS
EmergencyContact
- name: String
- phone: String
- relationship: String
+ EmergencyContact(name: String, phone: String)
+ EmergencyContact(name: String, phone: String, relationship: String)
RequestComment
- author: User
- text: String
- timeStamp: LocalDateTime
+ RequestComment(author: User, text: String)
+ RequestComment(author: User, text: String, timeStamp: LocalDateTime)
<<enumeration>>
UrgencyTier
LIFE_THREATENING
URGENT
STANDARD
Person
- id: UUID
- firstName: String
- lastName: String
- age: int
- needsMedicalAssistance: boolean
+ Person(firstName: String, lastName: String, age: int)
+ Person(id: UUID, firstName: String, lastName: String, age: int, needsMedicalAssistance: boolean)
+ needsMedicalAssistance(): boolean
+ getFullName(): String
Pet
- id: UUID
- name: String
- species: String
+ Pet(name: String, species: String)
+ Pet(id: UUID, name: String, species: String)
ShelterResource
- type: String
- quantity: int
- availableUntil: LocalDateTime
- lowStockThreshold: int
- unit: String
+ ShelterResource(type: String, quantity: int)
+ ShelterResource(type: String, quantity: int, availableUntil: LocalDateTime, lowStockThreshold: int, unit: String)
+ checkStockLevel(): boolean
+ getType(): String
Location
- addressLine1: String
- addressLine2: String
- city: String
- state: String
- zipCode: String
- latitude: double
- longitude: double
+ Location(latitude: double, longitude: double)
+ Location(addressLine1: String, addressLine2: String, city: String, state: String, zipCode: String)
+ Location(addressLine1: String, addressLine2: String, city: String, state: String, zipCode: String, latitude: double, longitude: double)
+ formatAddress(): String
+ toAddress(): Address